

Q1

Suppose $f_1(x) = x^3$ and $f_2(x) = x$ denote the profits of Company A and Company B, respectively, throughout 1 year (the beginning of the year is denoted by $x = 0$ and the ending denoted by $x = 1$). The predicted profits of Company A and Company B of the same year are given by the functions $g_1(x) = \sqrt{x}$ and $g_2(x) = e^x$, respectively. The curves represented by the functions f_1 and g_1 are shown in Figure M2W3G2, and the curves represented by the functions f_2 and g_2 are shown in Figure M2W3G3.

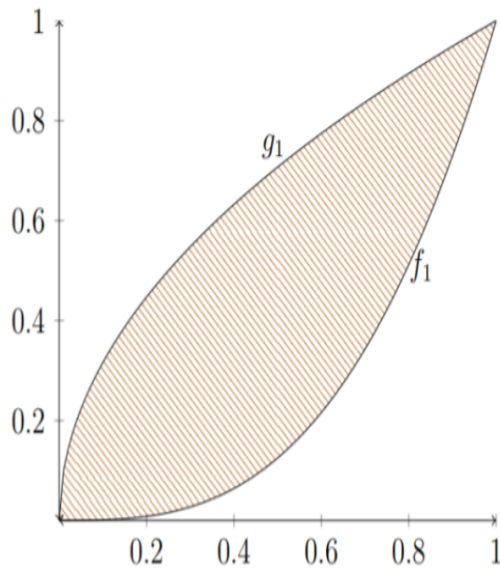


Figure: M2W3G2

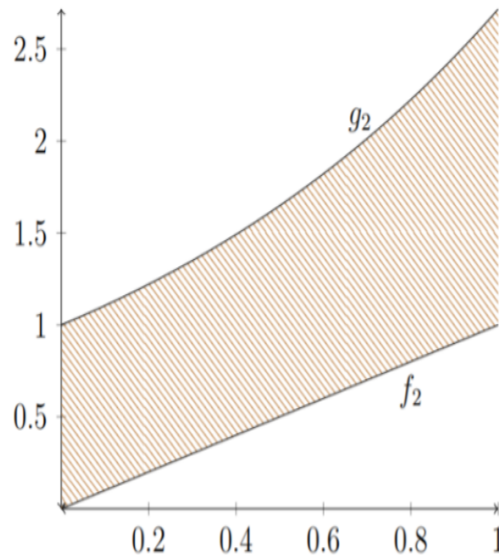


Figure: M2W3G3

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Suppose the area of the region bounded by the two curves (the original curve and the predicted curve) in the interval $[0,1]$ is defined to be the error in prediction.

What will be the difference between the minimum values of f_2 and g_2 in the interval $[0,1]$?



Solution:

- As both the functions are of increasing in the interval $[0,1]$, thus the minimum value will when $x=0$
 - $g_2(0) = e^0 = 1$ and $f_2(0) = 0$
 - Thus the difference between the minimum values of f_2 and g_2 in the interval $[0,1]$ is 1

Q2

Suppose $f_1(x) = x^3$ and $f_2(x) = x$ denote the profits of Company A and Company B, respectively, throughout 1 year (the beginning of the year is denoted by $x = 0$ and the ending denoted by $x = 1$). The predicted profits of Company A and Company B of the same year are given by the functions $g_1(x) = \sqrt{x}$ and $g_2(x) = e^x$, respectively. The curves represented by the functions f_1 and g_1 are shown in Figure M2W3G2, and the curves represented by the functions f_2 and g_2 are shown in Figure M2W3G3.

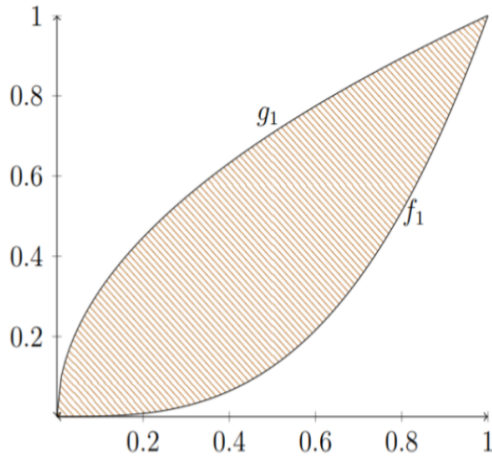


Figure: M2W3G2

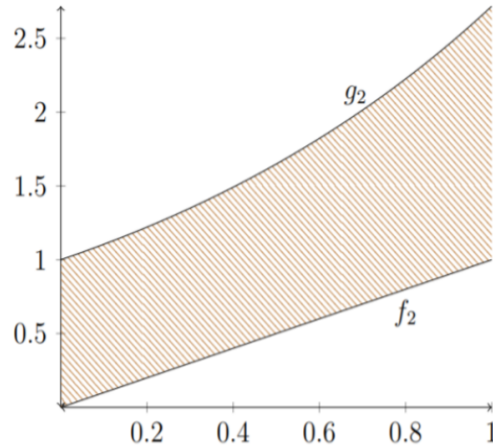


Figure: M2W3G3

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Suppose the area of the region bounded by the two curves (the original curve and the predicted curve) in the interval $[0, 1]$ is defined to be the error in prediction.

What will be error in prediction for Company A?



Solution:

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The error in prediction for Company A is the area enclosed by the functions f_1 and g_1 in the interval $[0, 1]$.

The area A_1 between the graph of f_1 and the interval $[0, 1]$ in the X -axis is,

$$\int_0^1 f_1(x) \, dx = \int_0^1 x^3 \, dx = \frac{x^4}{4} \Big|_0^1 = \frac{1}{4}.$$

The area A_2 between the graph of g_1 and the interval $[0, 1]$ in the X -axis is,

$$\int_0^1 g_1(x) \, dx = \int_0^1 \sqrt{x} \, dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \Big|_0^1 = \frac{2}{3}.$$

The area enclosed by the functions f_1 and g_1 in the interval $[0, 1]$ is $= A_2 - A_1 = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$.

Q3

Suppose $f_1(x) = x^3$ and $f_2(x) = x$ denote the profits of Company A and Company B, respectively, throughout 1 year (the beginning of the year is denoted by $x = 0$ and the ending denoted by $x = 1$). The predicted profits of Company A and Company B of the same year are given by the functions $g_1(x) = \sqrt{x}$ and $g_2(x) = e^x$, respectively. The curves represented by the functions f_1 and g_1 are shown in Figure M2W3G2, and the curves represented by the functions f_2 and g_2 are shown in Figure M2W3G3.

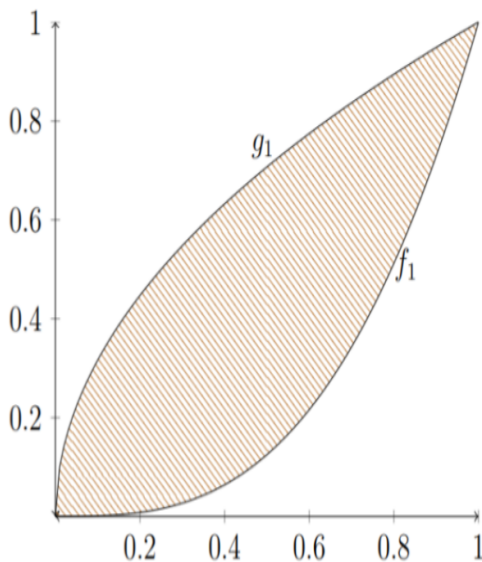


Figure: M2W3G2

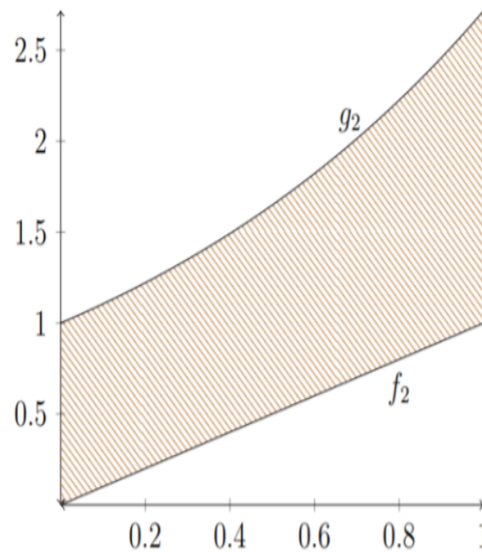


Figure: M2W3G3

Choose the correct options:

(a): The error in prediction for company A is more than that for company B.

(b): The error in prediction for company B is more than that for company A.

(c): The errors in prediction for both companies are equal.

(d): The error in prediction for Company A and Company B, cannot be compared using the given information.

Students choose an option

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Solution: The error in prediction for Company B is the area enclosed by the functions f_2 and g_2 in the interval $[0, 1]$.

The area A_3 between the graph of f_2 and the interval $[0, 1]$ in the X -axis is,

$$\int_0^1 f_2(x) \, dx = \int_0^1 x \, dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}.$$

The area A_4 between the graph of g_2 and the interval $[0, 1]$ in the X -axis is,

$$\int_0^1 g_2(x) \, dx = \int_0^1 e^x \, dx = e^x \Big|_0^1 = e - 1.$$

The area enclosed by the functions f_2 and g_2 in the interval $[0, 1]$ is $= A_4 - A_3 = e - 1 - \frac{1}{2} = e - \frac{3}{2}$.

Clearly, $e - \frac{3}{2} > \frac{5}{12}$.

Therefore, the error in prediction for company B is more than that for company A.

Q4

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Consider the function

$$f(x) = \frac{x^5}{5} - \frac{10x^4}{4} + \frac{35x^3}{3} - \frac{50x^2}{2} + 24x$$

Hint: $x^4 - 10x^3 + 35x^2 - 50x + 24 =$
 $(x-1)(x-2)(x-3)(x-4)$

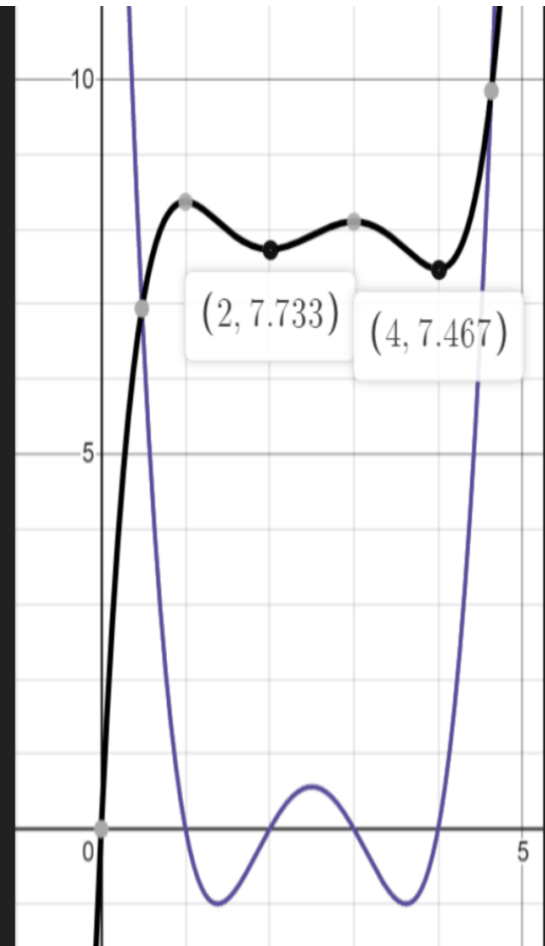
What is the local minimum value of 'f' attained at a critical point?



Solution:

$$f(x) = \frac{x^5}{5} - \frac{10x^4}{4} + \frac{35x^3}{3} - \frac{50x^2}{2} + 24x$$

- $f'(x) = x^4 - 10x^3 + 35x^2 - 50x + 24 = (x-1)(x-2)(x-3)(x-4) = 0$
- $x = 1, 2, 3, 4$
- Check for $f''(x)$
 - $f''(x) = 4x^3 - 30x^2 + 70x - 50$
 - $f''(1) = -6 < 0$ (point of maxima)
 - $f''(2) = 2 > 0$ (point of minima)
 - $f''(3) = -2 < 0$ (point of maxima)
 - $f''(4) = 6 > 0$ (point of minima)
- So $x = 2$, $f(2) = 7.73$ and $x = 4$, $f(4) = 7.46$
- So the local minimum value of 'f' attained at a critical point is **7.46**



$$x^4 - 10x^3 + 35x^2 - 50x + 24$$



$$\frac{x^5}{5} - \frac{10x^4}{4} + \frac{35x^3}{3} - \frac{50x^2}{2} + 24x$$

Q5

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Let $f(x) = 46x + \frac{22}{6}$, $0 \leq x \leq 6$. The estimated area obtained by dividing the interval into 3 sub-intervals of equal length and the left end points of the sub-intervals for height of the rectangles is (in square units)

Solution:

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Riemann sum:

Let $f : D \rightarrow \mathbb{R}$ be a function defined on some domain $D \subseteq \mathbb{R}$ and suppose the interval $[a, b]$ is contained in the domain D . Let P be a partition of the interval $[a, b]$ defined as $a = x_0 < x_1 < \dots < x_n = b$, $x_i^* \in [x_{i-1}, x_i]$. Define $\Delta x_i = x_i - x_{i-1}$,

The Riemann sum of f w.r.t. the above data is defined as $S(P) = \sum_{i=1}^n f(x_i^*) \Delta x_i$.

$$\Delta x_i = 2, \quad x_i^* = 0, 2, 4$$

$$S(P) = f(0) \cdot 2 = (46 \cdot 0 + 22/6) \cdot 2 = 7.33$$

$$S(P) = f(2) \cdot 2 = (46 \cdot 2 + 22/6) \cdot 2 = 191.33$$

$$S(P) = f(4) \cdot 2 = (46 \cdot 4 + 22/6) \cdot 2 = 375.33$$

$$\text{Riemann sum} = 7.33 + 191.33 + 375.33 = \sim 574$$

Q6

If $x - y = 1$, find the least value of $2xy$

Solution:

- $x - y = 1$ which implies $y = x - 1$
- $2xy = 2x(x-1) = 2x^2 - 2x$
- Consider $f(x) = 2x^2 - 2x$
- $f'(x) = 4x - 2 = 0$, which implies $x = 0.5$
- $f''(x) = 4 > 0$ (Point of minima)
- So the minimum value will be at $x = 0.5$ which is $f(0.5) = 2(0.5)^2 - 2(0.5) = -0.5$